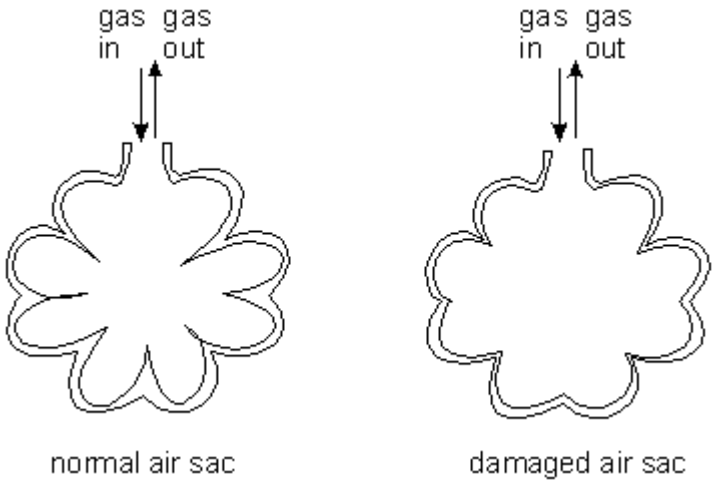
Q1. People who have emphysema have damaged air sacs in their lungs. The diagrams show a section through a normal air sac and a section through a damaged air sac.



(a) Gas exchange takes place at the inside surface of the air sac when a person breathes.

(i) Which **two** gases are exchanged at this surface of the air sac?

..... and .....

1 mark

(ii) The amount of gas exchanged is smaller in a damaged air sac.
Explain why.

.....

.....

1 mark

(b) The list shows four substances present in cigarette smoke.

carbon particles **carbon monoxide** **nicotine** **tar**

Choose from the list the substance which:

(i) causes addiction to smoking cigarettes;

.....

1 mark

(ii) may cause lung cancer;

.....

1 mark

(iii) is carried instead of oxygen in the red blood cells.

.....

1 mark

Maximum 5 marks

Mark schemes

Q1.

(a) (i) oxygen and carbon dioxide

1 (L5)

(ii) smaller surface area

1 (L5)

answers may be in either order
***both** are required for the mark*
accept 'O_2 and CO_2'
accept 'nitrogen' instead of oxygen and carbon dioxide
*do **not** accept 'smaller'*

(b) (i) nicotine

1 (L5)

(ii) tar

1 (L5)

(iii) carbon monoxide

1 (L5)

[5]